nervous system is being controlled, as irresistibly as if it were a helpless drug addict, or as if the cuckoo were a scientist plugging electrodes into its brain.

But even if we now feel more personal sympathy for the manipulated foster parent, we can still ask why natural selection has allowed the cuckoos to get away with it. Why haven't host nervous systems evolved resistance to the red gape drug? Maybe selection hasn't yet had time to do its work. Perhaps cuckoos have only in recent centuries started parasitizing their present hosts, and will in a few centuries be forced to give them up and victimize other species. There is some evidence to support this theory. But I can't help feeling that there must be more to it than that.

In the evolutionary 'arms race' between cuckoos and any host species, there is sort of built-in unfairness, resulting from unequal costs of failure. Each individual cuckoo nestling is descended from a long line of ancestral cuckoo nestlings, every single one of whom must have succeeded in manipulating its foster parent. Any cuckoo nestling that lost its hold, even momentarily, over its host would have died as a result. But each individual foster parent is descended from a long line of ancestors many of whom never encountered a cuckoo in their lives. And those that did have a cuckoo in their nest could have succumbed to it and still lived to rear another brood next season. The point is that there is an asymmetry in the cost of failure. Genes for failure to resist enslavement by cuckoos can easily be passed down the generations of robins or dunnocks. Genes for failure to enslave foster parents cannot be passed down the generations of cuckoos. This is what I meant by 'built-in unfairness', and by 'asymmetry in the cost of failure'. The point is summed up in one of Aesop's fables: 'The rabbit runs faster than the fox, because the rabbit is running for his life while the fox is only running for his dinner.' My colleague John Krebs and I have dubbed this the 'life/dinner principle'.

Because of the life/dinner principle, animals might at times behave in ways that are not in their own best interests, manipulated by some other animal. Actually, in a sense they are acting in their own best interests: the whole point of the life/dinner principle is that they theoretically could resist manipulation but it would be too costly to do so. Perhaps to resist manipulation by a cuckoo you need bigger eyes or a bigger brain, which would have overhead costs. Rivals with a genetic tendency to resist